Uniform asymptotic equivalence of second-order Zarankiewicz criteria

Nikita Lebedev

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Editorial note. One bundled analytic asymptotic-comparison object. Complete mathematical paper and original bibliography retained. Every native TeX input is recursively expanded with explicit line boundaries and locator-map.json; original main.bbl is inserted verbatim at the bibliography command, without reconstructing references or running external BibTeX. Original source/archive/input files preserved.

Editorial note. Original1211–1212 explicitly distinguishes written general density/two-column extension/transfer-graph proofs from exact-value computational certificates. Selected density and extension have their complete original written proofs; exact finite tables, six-column and odd-width certificate-dependent families are unselected context, not counted or independently regenerated/certified. External Zenodo evidence pointers and original author scope limitations remain in full context.

Editorial note. Original AI assistance statement1226–1229 says tools assisted exploratory computations, software development and draft preparation, with author responsibility for mathematical claims/exposition/data. This disclosure is retained without selected-core discovery inference, AI-free labeling, author certification or dependency-chain investigation.

Editorial note. Authors retain mathematical responsibility. No proof repair, added assumptions, independent mathematical review or counterexample search.

Allowed standard prerequisites (editorial).

- C4-free maximum bases, bilinear coefficient relations, recursive-line/signed closure, linear set-system matchings and fixed-width Zarankiewicz bounds. All original definitions, maximum-base hypotheses, dimension regimes and absolute versus width-dependent constants retained. Earlier incidence-subsequence and ordinary terminal-range results remain cited background.. Complete original written proof of the general density and two-column extension included; no regeneration of computational certificates or reconstruction of external prior results.

1 Introduction and main results

An ordinary C_4-free matrix can be augmented by pairing unused cells, each pair representing the square of a binomial in bilinear monomials. The problem is to maximize the displayed number of squares while ensuring that no representation by fewer bilinear squares exists. Löfberg and Qi introduced the second-order number z_2 and the sufficient recursive criteria defining z_{RL} and z_{SL} [8]. Their hierarchy is

$$z_{\mathrm{RL}}(m,n) \le z_{\mathrm{SL}}(m,n) \le z_2(m,n) \le \mathrm{BSR}(m,n). \tag{1}$$

The first three parameters require a maximum ordinary singleton base; BSR ranges over all SOS biquadratic forms. The author's earlier work [7] concerns augmentations under stronger compatibility conditions. Here we study the signed and recursive-line comparison posed in [3], together with its relation to z_2.

Three levels must be distinguished: acceptance of an individual display, the optimum on a prescribed maximum base, and the optimum when the base may also vary. The criteria can differ at the first two levels even where the optimized parameters agree. Section 5.2 explains the display-level obstruction, and Proposition 9.8 gives a fixed-base example. Our first result treats every maximum base. For such a base F, write $R(F)$ and $S(F)$ for its largest recursive-line and signed added-pair counts, with F fixed and the added pairs free.

Theorem 1.1 (Uniform half density)**.** *There is an absolute constant C such that every maximum $m \times n$ base F satisfies*

$$z(m,n) + R(F) \geq \frac{mn}{2} - C\max(m,n).$$

In particular, uniformly over dimensions and maximum bases as $\min(m,n) \to \infty$, both $R(F)$ and $S(F)$ equal $(1/2 + o(1))mn$, and

$$z_X(m,n) = \big(\tfrac{1}{2} + o(1)\big)mn \qquad (X \in \{\mathrm{RL}, \mathrm{SL}, 2\}).$$

For all dimensions, the same constant gives

$$S(F) - R(F) \leq \frac{z(m,n)}{2} + C\max(m,n), \qquad z_2(m,n) - z_{\mathrm{RL}}(m,n) \leq \frac{z(m,n)}{2} + C\max(m,n).$$

The construction organizes long column fans by a matching in a linear set system. Section 3 proves the theorem using that matching lemma and degree estimates for ordinary maximum bases. Along the odd-prime incidence subsequence, half density is already implicit in [8, Corollary 6.4]; the theorem covers every growing rectangle and every prescribed maximum base.

A separate extension treats fixed widths. Put

$$U(m,n) = \left\lfloor \frac{mn + z(m,n)}{2} \right\rfloor, \qquad U_n(m) = \left\lfloor \frac{(n+1)m + \binom{n}{2}}{2} \right\rfloor. \tag{2}$$

The expressions agree in the ordinary terminal range $m \geq \binom{n}{2}$ [4].

Theorem 1.2 (Fixed-width deficit)**.** *For every fixed $n \geq 3$ and $X \in \{\mathrm{RL}, \mathrm{SL}, 2\}$,*

$$z_X(m,n) = \frac{n+1}{2}m + O_n(1) \qquad (m \longrightarrow \infty). \tag{3}$$

More precisely, an explicit construction has $0 \leq U_n(m) - z_{\mathrm{RL}}(m,n) \leq n^3/6 + O(n^2)$ for all sufficiently large m at each width.

The cell cap gives the leading upper bound; the extension supplies the matching lower bound. The constant here depends on n. This fixed-width statement is distinct from Theorem 1.1, where both dimensions grow and the estimate is uniform.

Corollary 1.3 (Unrestricted asymptotic comparison)**.** *For positive integers m, n, uniformly as $\max(m,n) \to \infty$,*

$$\frac{z_{\mathrm{SL}}(m,n)}{z_{\mathrm{RL}}(m,n)} \longrightarrow 1, \qquad \frac{z_2(m,n)}{z_{\mathrm{RL}}(m,n)} \longrightarrow 1.$$

Proof. A counterexample sequence either has a subsequence with $\min(m,n) \to \infty$, contradicting Theorem 1.1, or has a subsequence with a fixed shorter side. The latter contradicts Theorem 1.2, after transposition; widths one and two have the known exact equalities [8]. □

The ratio limit is a leading-order statement, not exact equality. The extension in Section 4 allows arbitrary input displays and proves the fixed-width theorem without the specialized constructions used later in the paper.

For individual displays, Theorem 5.1 proves that both criteria are exact for one added pair on an inclusion-maximal base. Theorems 5.4 and 5.5 characterize the opposite end of the closure: once the signed closure identifies all selected pairs, signed acceptance is equivalent to irreducibility. The identification hypothesis cannot be dropped.

Our finite and constructive results give exact equality in the following ranges.

Theorem 1.4 (Complete six-column values)**.** *For every integer $m \geq 2$,*

$$z_{\mathrm{RL}}(m,6) = z_{\mathrm{SL}}(m,6) = z_2(m,6) = \begin{cases} 7, & m = 2, \\ 4m, & 3 \leq m \leq 7, \\ \lfloor (15m+15)/4 \rfloor, & 8 \leq m \leq 14, \\ \lfloor (7m+15)/2 \rfloor, & m \geq 15. \end{cases} \tag{4}$$

At $m = 1$ all six cells are fixed, so the common value is 6. The new actual upper bounds at 5×6, 6×6 and 7×6 are $20, 24, 28$. The recursive values and witnesses at the latter two sizes are already in [3]. At $8 \leq m \leq 16$, the constructions of [3] attain the cell cap. Section 7 supplies the all-length continuation, starting with their Figure 4 witness. Theorem 6.2 also gives $z_{\mathrm{RL}}(7,7) = z_{\mathrm{SL}}(7,7) = z_2(7,7) = 33$.

Theorem 9.3 proves

$$z_{\mathrm{RL}}(11,5) = z_{\mathrm{SL}}(11,5) = 37, \qquad z_{\mathrm{RL}}(8,7) = z_{\mathrm{SL}}(8,7) = 38.$$

These are signed upper bounds: they leave $z_2(11,5) \in [37,38]$ and $z_2(8,7) \in [38,39]$. The 11×5 equality, combined with the earlier finite values and the following constructions, implies $z_{\mathrm{RL}} = z_{\mathrm{SL}}$ whenever the shorter side is at most six, except possibly at 14×4 and its transpose (Corollary 9.7).

Theorem 1.5 (Odd-width exact formulas)**.** *All three parameters have the following values; the width-five formula is established in [13, Theorem 5.4 and Corollary 5.5].*

n	*row range*	$z_{\mathrm{RL}}(m,n) = z_{\mathrm{SL}}(m,n) = z_2(m,n)$
5	$m \geq 12$	$3m + 5$
7	$m \geq 19$	$4m + 10$
9	$m \geq 36$	$5m + 18$
11	$m \geq 53$	$6m + 27$.

The thresholds are least for all three parameters at widths seven, nine and eleven. At width five, twelve is least for z_{RL} and z_{SL}; whether $z_2(11,5) = 38$ remains open.

Section 8 presents a common one-row construction for widths seven, nine and eleven. Chen, Löfberg and Qi [1, Theorem 5.3] determine all three parameters on the incidence slice $m = \binom{n}{2}$ for the orders listed there, including $z_2(21,7) = 94$ and $z_2(36,9) = 198$. These agree with Theorem 1.5, which determines the parameters for all $m \geq 19$ at width seven and all $m \geq 36$ at width nine. At width eleven it gives $z_{\mathrm{RL}}(55,11) = z_{\mathrm{SL}}(55,11) = z_2(55,11) = 357$, confirming the $n = 11$ value in [1, Conjecture 4.10] by a different witness.

The uniform density and structural results have written proofs. The fixed-width argument starts from the previously established width-three and width-four families. Our exact-value results are computer-assisted: the finite exclusions use checked refutations or independent exhaustive enumerations, and the infinite exact families use symbolic certificate tables together with the lifting argument in Lemma 7.1. Appendix A identifies the data and verification methods.

2 Configurations, coefficient rules and ordinary bounds

Write $u_{rc} = x_r y_c$ for a cell of an $m \times n$ rectangle. A configuration consists of a singleton set F and a matching D of unordered pairs of distinct cells outside F. All selected supports are disjoint. Its displayed polynomial and length are

$$P_{F,D} = \sum_{a \in F} u_a^2 + \sum_{\{a,b\} \in D} (u_a + u_b)^2, \qquad N = |F| + |D|. \tag{5}$$

We call the display irreducible when its minimum real bilinear SOS length $\mathrm{sos}(P_{F,D})$ equals N. A *limited* configuration has a C_4-free singleton set of size $z(m,n)$. The definitions of z_2, z_{RL} and z_{SL} are those of [8, Definitions 3.1, 4.11, 7.3]. In particular, z_{RL} is exactly the parameter of [8, Definition 4.11]: its closure RW3$^+$ already includes the complementary-pair rule of [8, Definition 4.5]. Hole grounds are the unoccupied-companion case of the rectangle-transfer rule in [8, Definition 4.1], also used in [8, Lemma 4.8]. Transport along identified classes and simultaneous complementary identifications are rules of the same closure. Throughout, a maximum base has maximum cardinality $z(m,n)$; it is therefore also inclusion-maximal.

In any Gram representation of (5), let v_a be the coefficient vector at cell a. An occupied cell has $\|v_a\| = 1$, and a hole has $v_a = 0$. Put $\delta_{ab} = 1$ when $\{a,b\} \in D$ and zero otherwise. Distinct cells on one row or column satisfy $\langle v_a, v_b \rangle = \delta_{ab}$. For the two diagonals $\{a,b\}$ and $\{c,d\}$ of a genuine rectangle,

$$\langle v_a, v_b \rangle + \langle v_c, v_d \rangle = \delta_{ab} + \delta_{cd}. \tag{6}$$

A deviation means $\langle v_a, v_b \rangle - \delta_{ab}$. Accordingly, zero deviations on either diagonal transfer to the other. If a rectangle has a hole, its diagonal through that hole has zero inner product and zero prescription, so both deviations vanish. This hole ground follows directly from the coefficient equations. If $\delta_{ab} = 1$, its zero deviation identifies the two unit vectors. Two selected complementary diagonals are both identified: their deviations sum to zero and each is nonpositive by Cauchy–Schwarz. These are the line, rectangle, saturation and complementary-pair rules of RW3$^+$. A certificate is complete if all selected pairs are identified and the N distinct selected classes are pairwise orthogonal. It then forces rank at least N in every Gram representation.

The signed rule RW3$^\pm$ additionally considers the graph of class-pair deviations. Rectangle edges reverse signs. A component is forced to zero if it has a known zero, an odd sign cycle, or unresolved selected-pair inequalities in both parity classes. One then contracts newly identified pairs and repeats. We use its established soundness, not a converse. In particular, failure of either criterion is not itself reducibility.

Lemma 2.1 (Ordinary terminal bases and the cell cap)**.** *For $m \geq \binom{n}{2}$,*

$$z(m,n) = m + \binom{n}{2}. \tag{7}$$

Every extremal base in this range has one degree-two row for each column pair and otherwise degree-one rows. If $m < \binom{n}{2}$, then $z(m,n) \leq m + \binom{n}{2} - 1$. For every limited simple display, $N \leq U(m,n)$.

Proof. If d_r are the singleton row degrees, then

$$|F| - m = \sum_r (d_r - 1) \leq \sum_r \binom{d_r}{2} \leq \binom{n}{2}.$$

Equality requires each d_r to be one or two and every column pair to occur once. The indicated incidence rows and additional singleton rows attain the bound. If $m < \binom{n}{2}$, equality in the same upper bound would require $\binom{n}{2}$ degree-two rows, which is impossible. This is the classical terminal formula [4]; the equality description is also recorded in [14, Lemma 4.6]. If H is the number of holes, simplicity gives

$$|F| + 2|D| + H = mn, \qquad N = \frac{mn + |F| - H}{2}. \tag{8}$$

This proves the cap, as in [8, Proposition 3.2]. □

Lemma 2.2 (Ordinary six-column values)**.** *For $6 \le m \le 14$, the ordinary value is $z(m, 6) = \lfloor (3m + 15)/2 \rfloor$.*

Proof. Pair counting and $\binom{d}{2} \ge 2d - 3$ give the upper bound. For the lower bound put $t = \lfloor (15 - m)/2 \rfloor$ and take the first t triples in $012, 034, 135, 245$ as singleton row supports, with zero-based column indices. Their column pairs are disjoint. Add $m - t$ unused column-pair rows; there are enough since $m + 2t \le 15$. This C_4-free support has $3t + 2(m - t) = \lfloor (3m + 15)/2 \rfloor$ cells. □

Lemma 2.3 (Complete-square heredity)**.** *Every subfamily of the displayed squares of an irreducible SOS is irreducible. Deleting complete squares (singletons or pairs) also preserves* $\mathrm{RW3}^+$ *admissibility on the retained coefficient system. Deleting selected pairs, whose cells become holes, preserves* $\mathrm{RW3}^\pm$ *admissibility.*

Proof. A shorter SOS for the subfamily, together with the omitted squares, would shorten the full display. For the recursive assertion, replay a retained derivation in order. A removed companion cell becomes a hole; otherwise its rectangle prescription is unchanged. Identifications stay within complete selected supports. If one of two complementary selected pairs is removed, the other acquires a hole ground.

For $\mathrm{RW3}^\pm$, let $D' \subseteq D$ be the retained pairs. We show by induction over the rounds of the closure of D that its identifications of retained pairs, and its forced vertices between retained classes, are also obtained for D'. So assume this holds before some round.

Then the classes of D' on retained cells are at least as coarse as those of D. Map each vertex of D between retained classes to the vertex of D' containing it, or to nothing if its two classes have merged in D'; delete vertices involving a removed class. Under this map:

- an edge of D goes to an edge of D', or to a ground of D'. The latter happens when the other diagonal of its rectangle contains a removed cell, now a hole, or lies in a class merged in D';
- line grounds, hole grounds and identified-diagonal grounds persist.

Now take a forced component C of D. It breaks into pieces, each either the whole of C or joined within C to a deleted or merged vertex. A piece of the second kind is grounded in D'. The whole of C keeps its ground, its odd closed walk, or its two opposite selected inequalities, since those pairs are retained and unresolved in D'. So every forced vertex of D between retained classes is forced in D'. If opposite-parity vertices acquire the same image, their connecting path becomes an odd closed walk. Thus each retained image lies in a forced component of the new graph, and the induction proceeds.

At the terminal state of D, every retained pair is identified and every two retained classes are certified orthogonal, so D' satisfies $\mathrm{RW3}^\pm$. □

When restricting rows or columns, we delete every complete square that meets a deleted cell. A surviving half is not converted into a new singleton. A numerical limited bound on the retained rectangle is used only after verifying that its singleton base remains maximum.

3 Uniform half density

We first give a construction that retains an arbitrary C_4-free base. The only interaction condition concerns a matching of columns: no fixed row may contain two of its edges. A rainbow matching will supply this condition in the asymptotic argument.

Lemma 3.1 (Column-fan gluing)**.** *Let J be a matching of columns of a C_4-free base F, such that each row of F contains at most one edge of J. For $ab \in J$, let H_{ab} be the rows missing both a, b, and suppose $|H_{ab}| \geq 2$. There is at most one row z fixed in both columns. If it exists, color each row of H_{ab} by its common fixed column with z, giving separate colors to rows with no such column. Suppose H_{ab} can be ordered without equal consecutive colors. If no z exists, use any ordering. Then*

$$R(F) \geq \sum_{ab \in J} (|H_{ab}| - 1).$$

Here $R(F)$ denotes the pair optimum on this base even when F is not maximum.

Proof. Write an ordering as $q_0, \ldots, q_L$ and select $A_t = \{(q_t, a), (q_{t+1}, b)\}$ for $0 \leq t < L$. The two endpoint cells $(q_0, b), (q_L, a)$ are holes. Different fans use disjoint column sets. First normalize every chain backwards: the terminal a-hole starts the induction. Once A_{t+1} is identified, its representative (q_{t+2}, b) shares column b with the other companion (q_t, b) of A_t. This is a column ground, or a hole ground if that companion is absent. No product proved later is used.

For a foreign column, let Y_k denote its coefficient at q_k. A row grounds $A_t Y_k$ when $k = t, t+1$. When $k < t$, an a-rectangle transfers the target to $A_k Y_t$. When $k > t+1$, a b-rectangle gives $A_{k-1} Y_{t+1}$; this is a row ground for $k = t+2$, and otherwise an a-rectangle gives $A_{t+1} Y_{k-1}$. The gap decreases by two. Thus every pair is orthogonal to every foreign coefficient on its path.

Pairs of one fan share a column. For a pair P from another fan, the previous paragraph applies if one of its rows lies on this path. Otherwise both rows of P lie outside it, where columns a, b contain only fixed cells or holes. C_4-freeness supplies a hole in that two-by-two block; choosing the corresponding representatives grounds the product. All pair/pair products are therefore proved.

Consider a fixed cell (y, d). Path rows and own fan columns are already covered. If an exterior row y misses a or b, that cell is still a genuine hole: other fans use other columns. It directly grounds the target. Only the common fixed row z remains. For a fixed (z, d) outside a, b, the ordering gives an endpoint row r of the pair with $(r, d) \notin F$. Use the rectangle through this endpoint and (z, d). Its companion at (r, d) is a hole or a pair Q in another fan. In the latter case z misses that fan's other column, since z contains at most one edge of J. Since z is fixed in d, it lies outside that fan's path. Its missing cell in the other column is therefore a genuine hole. Use Q's representative in that column—available because all fans were normalized first—to ground its product with (z, a) or (z, b). This is a direct hole ground, so the common-root cases introduce no circular dependency.

Finally, a rectangle between two fixed cells has a nonfixed companion by C_4-freeness. Its deviation is zero by a hole or a pair/fixed or pair/pair product already proved. The two companions cannot be one selected pair: that pair's opposite corners lie on rows missing both of its columns and therefore cannot be the original two fixed cells. Line products are immediate. These are all the recursive-line rules, and every singleton has been retained. □

The coloring in the lemma is well-defined by C_4-freeness. A multiset can be ordered without equal consecutive colors whenever its largest color class has size at most half the total, rounded up: list the colors in decreasing multiplicity and fill alternate positions first,

then the remaining positions. Thus the lemma applies if every column incident with the common row z has degree $o(|H_{ab}|)$, uniformly over the selected edges.

3.1 Choosing column pairs

We use the following matching lemma. Its proof uses Tutte's theorem and the de Bruijn–Erdős incidence theorem.

Lemma 3.2 (Packing blocks in a linear set system)**.** *Let a set of N points be partitioned into parts $\mathcal{P}$, and let $\mathcal{L}$ be a family of subsets of size at least two. Distinct members of $\mathcal{L}$ meet in at most one point, and each meets every part in at most one point. For $q \geq 1$, there are q disjoint pairs, none inside a part and at most one inside each member of $\mathcal{L}$, if and only if*

$$N \geq 2q, \qquad |P| \leq N - q \ (P \in \mathcal{P}), \qquad |L| \leq N - q + 1 \ (L \in \mathcal{L}).$$

Proof. Necessity follows by counting points outside a part or a member of $\mathcal{L}$. First suppose $N = 2q$. For each L, add two adjacent vertices and join both to every point of L. Join two original points directly precisely when they lie in different parts and in no common member of $\mathcal{L}$. A perfect matching of this graph gives the required pairs: the two vertices of L either match each other or match two points of L. The intersection assumptions make these assignments unambiguous.

Apply Tutte's theorem [11]. Delete a vertex set S, containing h original points, both added vertices of r members of $\mathcal{L}$, and one added vertex of s others. Thus $|S| = h + 2r + s$. Components with no original point contribute at most s odd components. If the number c of components containing points is at most $h+2r$, Tutte's inequality follows. Assume $c > h+2r$.

If $c = 1$, then $h = r = 0$. For $s > 0$, every remaining added vertex still has a point neighbor, so there is at most one odd component, at most $|S|$. For $s = 0$, the graph has even order and its only point-containing component is even. If $r = 0$ and $c \geq 2$, any two points in different components must belong to the same part: a common member of $\mathcal{L}$ would still have an added vertex joining them. All remaining points therefore belong to one part, giving $c \leq 2q - h \leq q \leq h$, a contradiction.

It remains that $r \geq 1$ and $c \geq 3$. Choose one point from each component. Every pair of representatives belongs to exactly one part or one of the r removed members of $\mathcal{L}$. If the representatives are not all in one such set, the de Bruijn–Erdős incidence theorem [5] requires at least c sets. There are at most $r + \lfloor c/2 \rfloor$, since the parts are disjoint. This contradicts $c > h + 2r$. Thus every choice of representatives is contained in one set. Keeping two representatives and varying a third shows that all remaining points lie in that same set. A part gives $c \leq h$ as above. A member of $\mathcal{L}$ gives $2q-h \leq q+1$, hence $c \leq 2q-h \leq h+2 \leq h+2r$, again a contradiction. Tutte's theorem applies.

For general $N \geq 2q$, add $N-2q$ dummy points as one new part, meeting no member of $\mathcal{L}$. There are $2(N-q)$ points, and the balanced case applies with q replaced by $N-q$. Delete the pairs meeting dummy points. The remaining q pairs have the required properties. □

3.2 Below a fixed quadratic scale

The first range in Theorem 1.1. Orient $m \geq n$, put $M = |F| = z(m,n)$, and fix $\varepsilon = 1/100$, $c_0 = \varepsilon^2$. For all sufficiently large m, uniformly over $1 \leq n \leq m$,

$$M \geq (1-\varepsilon)n\sqrt{m}. \tag{9}$$

Indeed choose a prime $t = (1-o(1))\sqrt{m}$ with $t^2+t+1 \leq m$, and keep $\min(n, t^2+t+1)$ columns of its incidence matrix. The ratio of this support's size to $n\sqrt{m}$ tends to one uniformly: if

the plane has fewer than n columns, use $n \le m$. The prime number theorem and incidence construction suffice [6, 9].

Suppose $n \le m \le c_0 n^2$ and n is sufficiently large. Pair counting gives $M \le n\sqrt{m} + m \le 2\varepsilon n^2$. For row degrees d_r and $\mu = M/m$, using (9),

$$V := \sum_r (d_r - \mu)^2 \le n(n-1) + M - \frac{M^2}{m} \le 4\varepsilon n^2, \qquad \mu \le \sqrt{n} + 1 \le 2\varepsilon n.$$

Consequently $\max_r d_r \le (2\varepsilon + 2\sqrt{\varepsilon})n < n/2$. Apply Lemma 3.2 to the n columns, with singleton parts and with fixed-row neighborhoods of size at least two as $\mathcal{L}$. These sets are distinct and meet pairwise in at most one column, by C_4-freeness. With $q = \lfloor n/2 \rfloor$, all capacity conditions hold, including $d_r \le n - q + 1$. We obtain q disjoint column pairs with at most one inside each fixed row.

A column of degree E has at most $E + n - 1$ fixed cells in its incident rows. Applying pair counting to the remaining rows gives $M \le m + n - 1 + n\sqrt{m - E}$. Therefore

$$\sqrt{1 - E/m} \ge 1 - \varepsilon - \frac{\sqrt{m}}{n} - \frac{1}{\sqrt{m}} \ge 1 - 3\varepsilon, \qquad E \le 6\varepsilon m < m/5.$$

For every matched pair ab, its common-hole set has at least $3m/5$ rows, and every ordering color class has at most $m/5$ rows (including fresh singleton classes for large m). Lemma 3.1 applies. Writing e_a for a column degree, disjointness of the matched columns gives

$$M + R(F) \ge M + \sum_{ab \in J} (|H_{ab}| - 1) \ge |J|(m-1). \tag{10}$$

Here $|H_{ab}| = m - e_a - e_b + |N_F(a) \cap N_F(b)|$, and the sum of all matched column degrees is at most M. Thus $M + R(F) \ge \lfloor n/2 \rfloor (m-1) \ge mn/2 - m$. □

3.3 The quadratic and larger ranges

The following elementary replacement bounds row degrees when the number of rows is at least a fixed multiple of n^2.

Lemma 3.3. *For each $c > 0$, the row degrees of every maximum $m \times n$ base with $m \ge cn^2$ are bounded by a constant depending only on c, for all sufficiently large n.*

Proof. Pair counting gives $M/m \le 1 + 1/\sqrt{c}$. Choose an integer $K \ge 2(1 + 1/\sqrt{c})$. At least $m/2 \ge n$ rows have degree at most K, for sufficiently large n; these differ from the high row used below. Suppose a row has degree $a > 2\binom{K+1}{2}^2 + 1$ and put $k = K + 1$. Remove that row and a low-degree rows. Let A be its a columns; every surviving row meets A at most once.

There are $a + 1$ k-subsets of A with no repeated pair. Construct them greedily: after at most a choices there are at most $a\binom{k}{2}$ forbidden pairs, and a uniformly chosen k-subset contains fewer than one of them in expectation, since

$$\frac{a\binom{k}{2}^2}{\binom{a}{2}} < 1.$$

It therefore avoids every forbidden pair for some choice. Insert these $a + 1$ blocks in the freed rows. The result is C_4-free. At most $a + aK = ak$ cells were removed, but $(a+1)k$ were inserted, contradicting the maximum cardinality of F. □

Completion of Theorem 1.1. First assume $m \geq cn^2$ for fixed $c > 0$ and sufficiently large n. By Lemma 3.3, choose an integer $D \geq 2$, with $D = O_c(1)$, bounding all row degrees. Color each column pair by its unique common fixed row, or give it a fresh color when none exists. Set $\varepsilon = 1/5$. Call a column heavy if its degree exceeds εm. Since $M = O_c(m)$, there are $O_c(1)$ heavy columns. Omit these vertices from the column graph. Mark as forbidden each edge whose common fixed row meets a heavy column. For one heavy column h, C_4-freeness gives

$$\sum_{z \sim h} \binom{d_z - 1}{2} \leq \frac{D-2}{2} \sum_{z \sim h} (d_z - 1) \leq \frac{(D-2)(n-1)}{2}.$$

Thus only $O_c(n)$ edges are forbidden.

Omit one further vertex if necessary so the number $N = n - O_c(1)$ of remaining columns is even. This parity deletion does not enlarge the set of forbidden row colors. Choose a uniform perfect matching of the complete graph on these vertices, with $N \geq 4$. A given edge is selected with probability $1/(N-1)$, so the expected number of forbidden matching edges is $O_c(1)$. Two disjoint edges are selected together with probability $1/((N-1)(N-3))$. Each row color has at most $B = \binom{D}{2}$ edges, and

$$\sum_{\text{row colors } i} \binom{b_i}{2} \leq \frac{B-1}{2} \binom{n}{2} = O_c(n^2).$$

Thus the expected number of equal-color pairs of matching edges is $O_c(1)$; intersecting edges cannot both be selected, and fresh colors never collide. Some matching has total cost $O_c(1)$. Delete its forbidden edges and all but one edge of each repeated color. A color appearing r times costs at most $r - 1 \leq \binom{r}{2}$ deletions. We obtain a rainbow matching of $n/2 - O_c(1)$ edges.

For each retained matching edge ab, both columns have degree at most $m/5$, and their common fixed row, if any, meets no heavy column. Hence $|H_{ab}| \geq 3m/5$, while every ordering color class has size at most $m/5$. Lemma 3.1 and the exact summation (10) give

$$M + R(F) \geq |J|(m-1) \geq \frac{mn}{2} - O_c(m). \tag{11}$$

The constants depend only on c, uniformly in $m \geq cn^2$ and in F.

Take the fixed $c = c_0 = 10^{-4}$ from the first range. Together, the two arguments give $M + R(F) \geq mn/2 - Cm$ for some absolute C, for all $m \geq n \geq n_0$. Increasing C to at least $n_0/2$ covers $n < n_0$ as well, since the claimed lower bound is then nonpositive. Every line, rectangle and normalization rule is invariant under exchanging rows and columns, so transposition gives the stated bound for all dimensions.

The cell caps $S(F) \leq (mn - M)/2$ and $z_2(m,n) \leq (mn + M)/2$, together with $z_{\mathrm{RL}}(m,n) \geq M + R(F)$, give both comparison bounds. Finally $M/(mn) \leq 1/n + 1/\sqrt{m} \to 0$ when both dimensions grow. The lower bound just proved and the cell cap therefore give the uniform half-density limits. □

4 A universal two-column extension

The exact-width constructions in Sections 7 and 8 use special interfaces. The following extension works for an arbitrary simple input, at the cost of a bounded number of holes at each fixed width.

Theorem 4.1 (Short path extension)**.** *Let P be any simple display on $m \geq 1$ rows and $n \geq 1$ columns. Add columns A, B and pairs*

$$Q_k = \{(k-1, A), (k, B)\}, \qquad 1 \leq k \leq m-1,$$

leaving $(0,B),(m-1,A)$ *empty. The new polynomial* P^+ *satisfies* $\mathrm{sos}(P^+)=\mathrm{sos}(P)+m-1$. *Irreducibility,* $\mathrm{RW3}^+$ *and* $\mathrm{RW3}^\pm$ *each hold before the extension if and only if they hold afterward.*

If one also adds $2n+1$ *fresh two-singleton rows, one for each column pair involving* A *or* B*, let* f,d,N,H *count the singletons, pairs, displayed squares and holes. The exact changes are*

$$\begin{aligned} m' &= m+2n+1, & n' &= n+2,\\ f' &= f+4n+2, & d' &= d+m-1,\\ N' &= N+m+4n+1, & H' &= H+2n^2+n+2. \end{aligned} \tag{12}$$

The minimum SOS length increases by $m+4n+1$*. A maximum terminal singleton base remains maximum, and* $m'-\binom{n+2}{2}=m-\binom{n}{2}$.

Proof. For $m\geq 2$, the first pair normalizes through $(0,B)$, and the ordinary wave normalizes every Q_k. Distinct new classes share a column and are orthogonal. Consider any old coefficient $X_{l,c}$, including an unresolved half. A same-row ground handles $k=l$. Otherwise the B,c rectangle gives

$$\langle Q_k, X_{l,c}\rangle \longleftrightarrow \langle X_{k,c}, B_l\rangle.$$

If $l=0$ or $X_{k,c}$ is a hole, it is grounded. Otherwise $B_l=Q_l$, whose outgoing representative is $(l-1,A)$. If $k=l-1$, this is a row ground. In every other case a second rectangle gives

$$\langle Q_k, X_{l,c}\rangle \longleftrightarrow \langle A_k, X_{l-1,c}\rangle = \langle Q_{k+1}, X_{l-1,c}\rangle, \tag{13}$$

unless $k=m-1$, when A_k is a hole. The index move $(k,l)\mapsto(k+1,l-1)$ reaches a hole or a row ground. No old identification is assumed. Cases with coincident rows use the line rule instead. Thus the new chain is orthogonal to every old coefficient in every feasible Gram representation.

The additional singleton rows are sparse. A singleton in a new column is orthogonal to the chain by a common column. One in an old column uses the other new column, a hole in that row. Its products with old coefficients are line/hole grounds; the sole possible exception transfers to a just-established new-singleton/chain product. The new two-cell row supports are distinct, so their singleton set is C_4-free and their mutual products are likewise grounded.

After merging each new pair, every feasible Gram matrix is the direct sum of an old feasible block and an identity block of the asserted size. Conversely such a direct sum always represents the new polynomial. This proves the rank increments and irreducibility equivalence. All the new identities are $\mathrm{RW3}^+$ inferences. An old/old rectangle stays wholly old, and no new pair gives an old cell another representative; consequently new closure cannot repair an old uncertified component, for either criterion. The converses follow. The case $m=1$ has no chain pairs and is included. Finally the new singleton rows represent exactly the new column pairs, proving the maximum-base assertion and the counts. □

Proof of (3). Start at width $b=3$ for odd n, using the prior exact family $z_{\mathrm{RL}}(r,3)=2r$ for $r\geq 3$ [8]; it has three holes. For even n start at $b=4$, using [2] for $r\geq 15$, with zero or one hole. Repeated application of Theorem 4.1 reaches every target row count satisfying

$$m\geq\binom{n}{2}\quad(n\text{ odd}),\qquad m\geq\binom{n}{2}+9\quad(n\text{ even}).$$

The hole count is at most

$$h_b+\sum_{w=b,b+2,\ldots,n-2}(2w^2+w+2),\qquad h_3=3,\quad h_4=1. \tag{14}$$

It is independent of m and equals $n^3/3 + O(n^2)$. Equation (8) bounds the deficit from $U_n(m)$ by half this quantity, with the appropriate integer rounding. The prior cap and hierarchy prove (3) for all three parameters. □

The fixed-width gaps $z_2 - z_{\mathrm{RL}}$ and $z_{\mathrm{SL}} - z_{\mathrm{RL}}$ are consequently bounded as m grows, but the argument permits a nonzero bounded gap. The individual-configuration equivalence in Theorem 4.1 does not convert arbitrary optimized extremizers into one another.

5 Exactness at the first and final stages

5.1 One pair on a maximal singleton base

An inclusion-maximal C_4-free base need not be maximum. Nevertheless, maximality already makes the one-pair sufficient criteria exact.

Theorem 5.1 (One-pair completeness)**.** *Let F be inclusion-maximal and C_4-free in the full rectangle, and let $a, b \notin F$ be distinct. The display with singleton base F and one selected pair $\{a, b\}$ is irreducible if and only if it satisfies* RW3$^+$*, if and only if it satisfies* RW3$^\pm$*.*

Proof. Write $a = (r, c)$, $b = (s, d)$. On a C_4-free base, the unary recursive test has the following two geometric conditions:

(a) If the pair is nondegenerate, its companion cells $(r, d), (s, c)$ are not both fixed.
(b) There is no fixed cell (t, e), with $t \notin \{r, s\}$ and $e \notin \{c, d\}$, for which $(r, e), (s, e), (t, c), (t, d)$ are all fixed.

Repeated corners in a degenerate pair are counted only once. Under (a), the pair normalizes by a line or hole. Under (b), every pair/fixed product has a line or companion-hole ground. The remaining fixed/fixed products then follow from C_4-freeness and (6). Thus (a),(b) imply RW3$^+$.

Every failure has an actual shortening. In a nondegenerate fork set

$$E = x_r y_c + x_s y_d, \quad f = x_t y_e, \quad g = x_r y_e,\ h = x_t y_c,\ j = x_s y_e,\ k = x_t y_d.$$

These six displayed forms satisfy $Ef = gh + jk$, and

$$E^2 + f^2 + g^2 + h^2 + j^2 + k^2 = (E + f)^2 + (g - h)^2 + (j - k)^2. \tag{15}$$

For a degenerate fork there are five distinct forms with $Ef = g(h + k)$, and a four-square replacement is

$$\left(E + \frac{f}{\sqrt{2}}\right)^2 + \frac{f^2}{2} + \left(g - \frac{h + k}{\sqrt{2}}\right)^2 + \frac{(h - k)^2}{2}. \tag{16}$$

Suppose (a) fails. Let $u = (r, d)$, $v = (s, c)$ be fixed. Since $u_a u_b = u_u u_v$,

$$u_u^2 + u_v^2 + (u_a + u_b)^2 = u_a^2 + u_b^2 + (u_u + u_v)^2. \tag{17}$$

Put $F' = (F \setminus \{u, v\}) \cup \{a, b\}$. If F' contains a C_4, its four singleton squares reduce to two by the rectangle identity. Otherwise, maximality of F supplies a fixed rectangle completed by adding a. It must contain u or v, since otherwise it survives in F', and cannot contain both, since its fourth corner would be b. It therefore produces, possibly after transposition, a five-square subsum

$$(Au + Bv)^2 + (Av)^2 + (Bu)^2 + (Cu)^2 + (Cv)^2. \tag{18}$$

Here A, B, C are three row variables and u, v two column variables. Its explicit four-square replacement is

$$\left(Au + \frac{B+\sqrt{3}C}{2}v\right)^2 + \left(\frac{\sqrt{3}B-C}{2}v\right)^2 + \left(Av + \frac{B-\sqrt{3}C}{2}u\right)^2 + \left(\frac{\sqrt{3}B+C}{2}u\right)^2. \tag{19}$$

Expansion gives exactly (18). Every replacement is bilinear and shortens a genuine displayed subfamily. Thus failure of (a) or (b) is reducibility. Soundness RW3$^+$ $\Rightarrow$ RW3$^\pm$ $\Rightarrow$ irreducibility completes the equivalence. □

The strip obstruction itself is prior: the known two-column bound already implies that five displayed squares in a 3×2 strip are reducible [8, Appendix B.1]. Formula (19) makes the local certificate explicit. The additional point in Theorem 5.1 is that maximality forces one of the shortening configurations whenever unary normalization fails.

Corollary 5.2. *In an irreducible limited configuration, every individual selected pair, together with the entire singleton base, satisfies both unary tests.*

Proof. A maximum base is inclusion-maximal. Apply Lemma 2.3 and Theorem 5.1 to each one-pair subfamily. □

This is a necessary preprocessing condition, not joint sufficiency for several pairs. The maximality hypothesis also cannot be dropped: on 2×2, the three-square display $(x_1y_1 + x_2y_2)^2 + (x_1y_2)^2 + (x_2y_1)^2$ is irreducible but fails the signed test on its nonmaximal two-cell singleton base. Its Gram matrices have blocks $\left(\begin{smallmatrix} 1 & t \\ t & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 1-t \\ 1-t & 1 \end{smallmatrix}\right)$; positivity is $0 \le t \le 1$, with minimum rank three.

5.2 The final transfer graph

This section makes precise how the two criteria relate once every selected pair has been identified. We show that, from that point on, the orthogonality phase of RW3$^\pm$ is exact: it fails exactly when the displayed forms satisfy a nontrivial quadratic relation. We use only simplicity: no cell lies in two displayed squares. The base need not be C_4-free or maximum. The rules of RW3$^+$ and RW3$^\pm$, and their soundness proofs in [8], use only simplicity, although they are stated there for limited graphs.

5.2.1 The graph and its relations

Let (F, D) be a simple display. Its *classes* are the displayed squares' supports: the singletons of F and the pairs of D. For a class X put $f_X = \sum_{a \in X} u_a$. Thus

$$P_{F,D} = \sum_X f_X^2 \qquad \text{and} \qquad N = |F| + |D|.$$

Definition 5.3. The *final transfer graph* $\Gamma(D)$ is defined as follows.

- *Vertices* are the unordered pairs $\{X, Y\}$ of distinct classes. A *cross pair* of $\{X, Y\}$ is a pair of cells $x \in X$, $y \in Y$.
- A vertex is *grounded* if some cross pair
  (i) lies in one row or one column, or

(ii) spans a genuine rectangle whose other diagonal contains a hole or is a pair of D.

- *Edges*: each genuine rectangle with all four corners occupied, whose two diagonals are cross pairs of vertices v and w, gives an edge vw. It is a loop when $v = w$.

A component is *grounded* if it contains a grounded vertex. It is *bipartite* if it has no closed walk of odd length.

Every loop is at a grounded vertex. Suppose the two diagonals $\{(r,c),(s,d)\}$ and $\{(r,d),(s,c)\}$ of one rectangle give the same vertex. Then two corners in one row, or in one column, belong to the same class. Another cross pair of that vertex then lies in one line, so the vertex is grounded by (i).

Theorem 5.4. *Let $C = (C_{XY})$ be a real symmetric matrix indexed by classes. Then $\sum_{X,Y} C_{XY} f_X f_Y$ vanishes identically if and only if C has zero diagonal and the function $h(\{X,Y\}) = C_{XY}$ satisfies:*

- *h vanishes at grounded vertices;*
- *$h(v) = -h(w)$ along every edge vw.*

The space of such relations therefore has dimension equal to the number of ungrounded bipartite components of $\Gamma(D)$.

If such a component exists, then $\mathrm{sos}(P_{F,D}) \le N - 1$.

Proof. We compare coefficients of the biquadratic monomials.

- *Squares u_a^2.* This monomial occurs only in f_X^2 for the class X of a, with coefficient C_{XX}. Hence $C_{XX} = 0$.
- *Line monomials $x_r^2 y_c y_d$ and $x_r x_s y_c^2$.* Each arises from a single cell pair, and when its cells lie in distinct classes the coefficient is $2h(v)$. This is condition (i).
- *Rectangle monomials $x_r x_s y_c y_d$.* Each arises exactly from the two diagonals of one rectangle. A diagonal contributes $2h$ of its vertex when both cells are occupied and in distinct classes, and nothing otherwise (a hole, or a single class, where $C_{XX} = 0$). So the coefficient vanishes exactly when $h(v) + h(w) = 0$ along an edge, or $h(v) = 0$ under ground (ii).

On a component, these conditions force $h \equiv 0$ unless the component is ungrounded and bipartite. On such a component the solutions are the multiples of its ± 1 colouring.

Now take such a colouring H. It is symmetric, nonzero, has zero diagonal and trace, and satisfies $f^{\mathsf{T}} H f \equiv 0$. With $\lambda = \lambda_{\max}(H) > 0$, the matrix $W = I - H/\lambda$ is positive semidefinite of rank at most $N-1$, and $P_{F,D} = f^{\mathsf{T}} W f$. Factoring W gives at most $N-1$ bilinear squares; this is the rank step of [14, Lemma 2.13]. □

5.2.2 Completeness after identification

In any SOS representation of $P_{F,D}$, the coefficient equations of Section 2 hold for the cell vectors. Every rule of RW3$^+$ and RW3$^\pm$ is a deduction from these equations, together with the Cauchy–Schwarz inequality for the unit vectors of a selected pair. We say that a criterion *identifies D* if its closure resolves every selected pair.

Theorem 5.5. *Let (F,D) be simple.*

(a) If (F,D) satisfies RW3$^\pm$*, then $\Gamma(D)$ has no ungrounded bipartite component.*

(b) If RW3$^\pm$ *identifies D and $\Gamma(D)$ has no ungrounded bipartite component, then (F,D) satisfies* RW3$^\pm$*.*

(c) If (F,D) satisfies RW3$^+$*, then every component of $\Gamma(D)$ is grounded. Conversely, if* RW3$^+$ *identifies D and every component is grounded, then (F,D) satisfies* RW3$^+$*.*

In particular, if RW3$^\pm$ *identifies D, the following are equivalent:*

- *(F,D) satisfies* RW3$^\pm$*;*

- $\Gamma(D)$ *has no ungrounded bipartite component;*
- $P_{F,D}$ *is irreducible.*

Proof. (a). Let K be an ungrounded bipartite component with colouring H, and take $0 < t \le 1/\lambda_{\max}(H)$. Then $I - tH$ is positive semidefinite with unit diagonal; write $I - tH = B^{\mathsf{T}}B$ with columns w_X. Put $v_a = w_X$ for each $a \in X$, and $v_h = 0$ at holes. Since $f^{\mathsf{T}}Hf \equiv 0$, these vectors come from the SOS representation

$$P_{F,D} = \sum_i \big(\textstyle\sum_a (v_a)_i u_a\big)^2.$$

For $x \in X$ and $y \in Y$ with $\{X, Y\} \in K$, we get $\langle v_x, v_y\rangle = -tH_{XY} \neq 0$. But an accepting closure certifies $\langle v_x, v_y\rangle = 0$ in every representation.

(b). Once every pair is resolved, the current classes are the displayed supports. The signed transfer graph of that stage then contains all grounds and edges of $\Gamma(D)$:

- the line rule gives ground (i);
- transfer from a hole, or from an identified pair's diagonal, gives ground (ii);
- the rectangle identities give the sign-reversing edges, since all prescriptions between distinct final classes vanish.

Grounded components are forced by propagation, and every other component has an odd cycle. So every pair of classes is certified orthogonal.

(c). Every certified orthogonality $p \perp q$, with p, q in distinct final classes X, Y, lies in a grounded component of $\Gamma(D)$. We prove this by induction over the RW3$^+$ derivation.

- *Line rule.* This gives ground (i).
- *Saturation.* This stays within the same vertex, because identifications merge only the two cells of a selected pair.
- *Complementary-pair rule.* This only identifies pairs and produces no orthogonality.
- *Rectangle transfer* from the other diagonal $\{r, s\}$.
  - If that diagonal contains a hole, or is an identified pair, we get ground (ii).
  - If $r \perp s$ was certified, then $\delta(\{r, s\}) = 0$, so $\{r, s\}$ is not a selected pair and r, s lie in distinct classes. The rectangle is an edge to their vertex, which lies in a grounded component by induction.

Acceptance certifies every vertex. The converse is as in (b), without odd cycles. Saturation moves a certificate between the cross pairs of one vertex, and rectangle transfer carries it along an edge.

For the final equivalence: (a) and (b) give the first equivalence; Theorem 5.4 and the soundness of RW3$^{\pm}$ [8, Theorem 7.2] give the second. □

Remark 5.6. *The identification hypothesis is necessary.* Consider the 4×4 display T of [14, (40)]: the four diagonal cells as singletons, plus the six pairs F^+ of [14, (38)].

- A direct check shows that all 15 components of Γ are grounded. By Theorem 5.4 there is no product relation, in agreement with [14, Proposition 5.4].
- The display has no holes, no degenerate pairs and no complementary pairs, so neither closure identifies any pair.
- Yet T is a sum of nine squares [14, Theorem 5.3].

As Xu and Yan remark after their Proposition 5.4, shorter representations may use forms outside the displayed span. Identification is what excludes them here.

Related work. The proof of [14, Proposition 5.4] already observes that relations have zero diagonal and correspond to the kernel of the product map for one display. [8, Remark 7.8] describes the Gram perturbation mechanism. In their formulation, [1, Proposition 2.3] shows that after an identification prefix, grounding of every component suffices for RW3$^+$; this is

the converse half of Theorem 5.5(c). Here we add the necessity direction, the description of all product relations (Theorem 5.4), and the equivalence of RW3$^{\pm}$ with irreducibility when the signed closure identifies D.

Relation to [8, Remark 7.8]. That remark declines a converse for ungrounded bipartite signed components. The remark remains appropriate whenever identification is incomplete, including at fixed points with unresolved pairs. Theorem 5.4 gives the converse at the final round.

Consequence for the comparison. If RW3^{+} identifies D, then (F, D) satisfies RW3$^{\pm}$ but not RW3^{+} exactly when $\Gamma(D)$ has an ungrounded odd component and no ungrounded bipartite one. The 15×6 witness of [8, Theorem 7.5] and the display of Proposition 5.7 are of this kind; the latter's three odd components have sizes 5, 13 and 3. In any other signed-only display, RW3^{+} fails to identify some pair.

Before any identification. Consider a hole-free display. Before any identification, every current class has one cell, so each vertex has a single cross pair and at most one rectangle. The only pairs either rule can then resolve are degenerate and complementary pairs, and the two closures agree.

5.3 A signed-only full display

Proposition 5.7 (Full occupancy does not make the criteria agree)**.** *There is a full-support* 16×4 *display of length* $43 = U(16, 4)$ *on a maximum singleton base which satisfies* RW3$^{\pm}$ *but not* RW3^{+}*.*

Proof. Number cells by $a = 4r + c$, with indices starting at zero. The singleton row supports are (writing 01 for $\{0, 1\}$, and similarly)

$$01, 02, 03, 12, 13, 23, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3.$$

Take the following unordered pairs:

$$(2, 3), (5, 60), (7, 49), (9, 62), (10, 57), (12, 21), (15, 29),$$
$$(16, 50), (18, 61), (24, 31), (25, 48), (27, 28), (32, 38), (33, 40),$$
$$(34, 36), (37, 46), (41, 56), (44, 45), (20, 54), (42, 52), (53, 58).$$

The 22 singletons form a maximum base by Lemma 2.1. Every other cell occurs exactly once, so the length is $43 = U(16, 4)$.

Here the distinction is entirely in orthogonality: every selected pair normalizes recursively. Packet F16 gives a 31-step normalization sequence and a checker. After these identifications, form the graph on the $\binom{43}{2} = 903$ products of distinct selected classes. Line products are grounded; rectangle equations reverse signs along edges. A direct inventory gives exactly three components without a zero ground, of sizes $5, 13, 3$. Each contains a triangle. For example, writing $A = [53, 58]$, $B = [42, 52]$, $C = [41, 56]$ for three paired classes and $[a]$ for a singleton class, one triangle has vertices

$$[0]A, \qquad [1]B, \qquad [2, 3]C.$$

Its rectangle witnesses are $(0, 13; 0, 1)$, $(0, 10; 1, 2)$ and $(0, 14; 0, 2)$, where $(r, s; c, d)$ denotes rows r, s and columns c, d. Each product denotes the inner product of the corresponding class vectors. The signed rule grounds all three odd components. The recursive rules cannot enter them. The F16 reader reconstructs the entire product graph and checks all three triangles, proving the two outcomes. □

This example does not separate the optimized values, even on this base. Replacing $(20, 54), (42, 52), (53, 58)$ by $(20, 42), (52, 58), (53, 54)$ recovers the recursive length-43 construction of [2, Figure 2]. Thus full occupancy does not justify keeping the pairing unchanged when converting a signed display to a recursive one.

6 Finite SOS upper bounds

Theorem 6.1. *For these three rectangles,* $z_{\mathrm{RL}} = z_{\mathrm{SL}} = z_2$*, with*

$$z_2(5, 6) = 20, \qquad z_2(6, 6) = 24, \qquad z_2(7, 6) = 28.$$

We describe the three logically separate parts of the proof: coverage of the mathematical domain, validity of every reduction, and the final propositional contradiction. Full coordinate and clause data are kept in the companion files indexed in Appendix A.

6.1 Maximum bases and lower witnesses

Pair counting and $\binom{d}{2} \geq 2d - 3$ give ordinary maxima $14, 16, 18$. The first is most easily classified after transposition to 6×5. Equality forces two degree-three rows and four degree-two rows. The triples meet once, say $\{a, b, c\}$ and $\{a, d, e\}$; the remaining rows are exactly bd, be, cd, ce. There are $5 \cdot 3 = 15$ row-unordered bases on labeled columns, all isomorphic.

At 6×6, the only pair-count-compatible profiles at total 16 are $3^4 2^2$, $4\,3^2 2^3$, and $3^5 1$. The last is impossible because each column lies in at most two triple rows. In the middle profile, both triples must contain the two columns outside the degree-four row, creating a C_4. For $3^4 2^2$, every column lies in exactly two of the four triples. The columns identify with the six edges of K_4 and the triple rows with its stars. The two pair rows choose two of its three opposite-edge pairs. This gives $30 \cdot 3 = 90$ row-unordered labeled bases and one isomorphism class.

At 7×6, equality forces $3^4 2^3$. The same four triple rows occur, and all three opposite-edge pairs are required. There are 30 row-unordered labeled bases, again one isomorphism class. Exhaustive row-mask enumeration and all column permutations independently check these classifications and provide maps to the three canonical inputs.

The lower displays have $(|F|, |D|, H) = (14, 6, 4), (16, 8, 4), (18, 10, 4)$. The last two are the published Figure 2 witnesses of [3]; the first is an explicit additional construction. Actual-cell certificates use respectively 227, 345 and 468 inferences and certify all 190, 276 and 378 pairs of selected classes. Thus their displayed lengths are 20, 24 and 28 by soundness.

The cell caps are 22, 26 and 30. By complete-square heredity, excluding irreducible totals 21, 25 and 29 also excludes the respective next total: delete one complete pair square from a hypothetical irreducible display. The maximum singleton base remains unchanged.

6.2 Exact shortening certificates

The input data enumerate every unary choice and every disjoint binary choice not already excluded. Theorem 5.1 explains why unary filtering on a maximum base can be exact. The frozen exclusions additionally carry explicit polynomial identities, rather than relying on a Boolean criterion label. Retained choices need not all be irreducible; completeness requires that no irreducible choice be removed without proof.

We use explicit identities and coefficient-preserving rank reductions. First, an explicit SOS with fewer squares is checked by expanding every quartic coefficient over $\mathbb{Z}$ or $\mathbb{Q}(\sqrt{3})$. Second, a nonzero zero-diagonal symmetric matrix H with $f^T H f = 0$, where f is the vector

of displayed forms, gives a rank reduction. Since $\operatorname{tr} H = 0$, its largest eigenvalue λ is positive, and $I - H/\lambda$ is singular PSD. Thus $f^T(I - H/\lambda)f = f^T f$ has fewer squares. This uses an exact coefficient identity, not a numerical eigenvalue decision.

Fourteen further reducible subdisplays have explicit positive-rational-weight identities of the form

$$\sum_{j=1}^{N} f_j^2 = \sum_{i=1}^{N-1} \alpha_i \ell_i^2, \qquad \alpha_i \in \mathbb{Q}_{>0}, \quad \ell_i \in \mathbb{Q}[x_1, \dots, x_m, y_1, \dots, y_n]_{(1,1)}. \tag{20}$$

Here the subscript denotes bilinear forms. Clearing denominators reduces verification to integer coefficient expansion; each $\sqrt{\alpha_i}\ell_i$ is a real bilinear form. There is one additional core at 5×6, three at 6×6 and ten at 7×6. Each identity names the complete original singleton and pair supports, so a source hash and a checked coordinate map identify exactly which reducible subfamily is excluded. The accompanying explicit-core manifest binds all fourteen identities to these original sources.

A literal ten-to-nine reduction. For the first additional 6×6 core, the reduction also has a simple explicit form. Write $w_j = x_{\lfloor j/6 \rfloor} y_{j \bmod 6}$ with zero-based indices, put $p = w_{12} + w_{35}$, and let

$$L = \sum_{j \in \{0,2,17,20,23,24,29\}} w_j^2 + (w_5 + w_{32})^2 + p^2 + (w_{14} + w_{18})^2.$$

The rectangle identities give

$$\begin{aligned} 2L = {} & (w_0 - w_{14} + w_{23})^2 + (w_0 - w_{29})^2 + (w_5 - w_{14} - w_{18})^2 \\ & + (w_5 + w_{24} + w_{32})^2 + (w_{29} - w_{18})^2 + (w_{32} - w_{23} - w_{24})^2 \\ & + (w_2 - w_{20})^2 + (w_2 + p + w_{17} + w_{20})^2 + (w_{17} - p)^2. \end{aligned} \tag{21}$$

Dividing these nine forms by $\sqrt{2}$ shortens the ten-square display L. Restoring the nine singleton squares at $1, 6, 9, 10, 13, 15, 22, 31, 34$ gives an explicit 19-to-18 reduction of the same core. Its verification needs only integer coefficient expansion.

Restoring the omitted displayed squares transports a shortening of a complete subfamily to every completion containing it. Row/column substitutions are checked on the full polynomial. A local square 6×6 identity may also be transposed before embedding in six rows of a 7×6 rectangle; this is not a transposition of the whole nonsquare board.

6.3 Guarded row deletion and propositional coverage

The 7×6 canonical base has three degree-two rows. Deleting one leaves a maximum 6×6 base with 16 singleton cells; deleting two leaves a maximum 5×6 base with 14. Let t count distinct selected pairs touching the deleted union. From a hypothetical irreducible 29-square display, the retained lengths are $27 - t$ or $25 - t$. The already proved smaller cases give

$$t \geq 3 \quad \text{or} \quad t \geq 5, \qquad \text{respectively.} \tag{22}$$

A pair meeting both deleted rows is counted once. If H is the number of holes in the union and I the number of pairs wholly inside it, cell coverage gives $t = 4 - H - I$ or $t = 8 - H - I$. Therefore the respective constraints are $H + I \leq 1$ and $H + I \leq 3$. Every retained singleton base and its embedding is checked; deleting arbitrary degree-three rows would not justify the same limited bounds.

For each target, a Boolean variable chooses each allowed pair and one variable marks each possible hole. Pair conflicts enforce disjointness; cell clauses enforce either a hole or

an incident selected pair, and pair/hole conflicts exclude both. All negative triples of hole variables, together with a positive clause omitting each hole variable in turn, express exactly two holes. Hence the selected pair count is forced to be seven, nine or eleven. A verified reducible subfamily contributes the clause that not all its pairs may be chosen.

Rectangle	Target	Pair variables	Hole variables	Clauses	Additional explicit cores
5×6	21	58	16	1,416	1
6×6	25	90	20	3,070	3
7×6	29	138	24	8,395	10

There are no auxiliary variables or symmetry restrictions on the matching search. Symmetries are used only to transport valid reductions. The 7×6 total comprises 4,526 matching clauses, 609 guarded deletion clauses and 3,260 reduction images.[1] The mathematical input check reconstructs every clause. Separate DRAT certificates are then checked against the exact final formulas [12]. Thus every irreducible target would give a satisfying assignment, whereas each formula is unsatisfiable. This proves the required three exclusions and, with the base and lower certificates, Theorem 6.1.

6.4 The seven-by-seven square

Theorem 6.2. $z_{\mathrm{RL}}(7,7) = z_{\mathrm{SL}}(7,7) = z_2(7,7) = 33$.

Proof. Pair counting and $\binom{d}{2} \geq 2d - 3$ give $|F| \leq 21$. Equality forces seven degree-three rows covering every column pair once. Up to column relabeling, the triples through column 0 are $012, 034, 056$. The remaining triples through column 1 may then be taken as $135, 146$, forcing the remaining triples $236, 245$. Thus the Fano base is the unique maximum base up to row and column permutations. A 21-singleton, 12-pair witness on this base has an actual-cell recursive certificate, giving the lower value 33.

A target of 34 squares has 13 pairs and two holes among the 28 free cells. Of the 378 possible individual pairs, the 210 omitted choices have verified SOS shortenings. The complete matching formula represents all remaining choices with 168 pair variables and 28 hole variables. The additional cuts use exact polynomial reductions and verified row/column substitutions. There is also an acyclic use of $z_2(6,6) = 24$: deleting the row and column of any fixed incidence leaves a maximum 16-cell base. Retain exactly the displayed squares whose complete supports avoid that row and column. This sub-sum is irreducible by heredity. If T distinct selected pairs meet the deleted region, its length is $29 - T$, so $T \geq 5$. Only these incidence deletions are used; a nonincident deletion would leave a smaller singleton base and would not justify the same bound.

The complete 196-variable, 15,666-clause formula is reconstructed from these conditions, without symmetry restrictions on matchings, and has a separately checked DRAT refutation. The ordinary classification transports this exclusion to every maximum base. The cell cap is 35, and complete-square heredity excludes a hypothetical irreducible 35-square display by deleting one pair square. Hence $z_2(7,7) \leq 33$, completing the proof. The source coordinates, the reductions, input audit and proof check are indexed in packet F7. □

[1] These are 8,395 clause lines and 8,383 distinct clauses. Twelve hole/pair conflicts are repeated between the matching and deletion lists; repetitions do not affect satisfiability.

6.5 Further four- and five-column upper bounds

Theorem 6.3. *The three parameters have the following common values:*

(m,n)	$(7,5)$	$(8,5)$	$(10,5)$	$(9,4)$	$(10,4)$
$z_{\mathrm{RL}}(m,n) = z_{\mathrm{SL}}(m,n) = z_2(m,n)$	24	27	34	24	27.

Proof. Since $z_{\mathrm{RL}} \le z_{\mathrm{SL}} \le z_2$, it suffices to give recursive-line lower displays and upper bounds for z_2. The actual upper bounds use the complete-matching method of this section: each excluded local pattern has a shorter real SOS, while inherited bounds are applied only to complete selected squares on a maximum base of the smaller rectangle. No rejected recursive or signed certificate is an upper-bound premise. Each target exceeds the asserted value by one:

Rectangle	Maximum bases	$\lvert F\rvert$	Target pairs	Target holes	Target length
7×5	4	15	10	0	25
8×5	1	17	11	1	28
10×5	1	20	15	0	35
9×4	3	15	10	1	25
10×4	5	16	12	0	28

Here "maximum bases" counts row/column isomorphism classes. Their coverage is elementary. On five columns, pair counting gives the upper bounds $15, 17, 20$. At seven rows, either two triple rows use all three pairs in each triple, leaving four pair rows and one singleton row, or one triple is accompanied by six of the seven remaining column pairs. The triples in the first case meet in one column; the singleton is anchored either there or in an outer column. In the second case the omitted pair lies either outside the triple or crosses it. These are four types, with 145 row multisets on labeled columns in total. At eight rows the base is one triple and all seven remaining pair rows; at ten rows it is all ten pair rows. Larger row degrees are excluded by the same pair budget and C_4-freeness.

For four columns every maximum base consists of the six pair rows and singleton rows. At nine rows the anchor multiplicities are $(3,0,0,0)$, $(2,1,0,0)$ and $(1,1,1,0)$. At ten rows they are $(4,0,0,0)$, $(3,1,0,0)$, $(2,2,0,0)$, $(2,1,1,0)$ and $(1,1,1,1)$. Thus the table covers every maximum base.

For each type, every pairing with the stated hole count either appears in the matching domain or is excluded by a verified local reduction. The local exclusions are checked by exact coefficient identities or certified lower-rank real Gram matrices. The identities use integers, positive rational weights or $\mathbb{Q}(\sqrt{3})$. For the real Gram matrices, exact rational bounds keep the principal block positive definite and make an invertible-preconditioned Schur map a strict contraction of a closed box into itself. Its fixed point has zero Schur complement, giving a positive-semidefinite matrix of the smaller principal rank. The checks use exact LDL decompositions and rational arithmetic; rationality of the final matrix entries is not assumed. The maximum-retaining seven-row, five-column deletion clauses use $z_2(6,5) = 20$ (Theorem 6.1, transposed). The four-column inputs also use $z_2(6,4) = 16$ from [8, Corollary 4.23] and $z_2(7,4) = 19$, $z_2(8,4) = 21$ from [14, Theorems 4.2 and 4.4]; the ten-row input uses the nine-row bound only after that separate proof is complete. Only the 10×5 formula uses lexicographic constraints: all 120 column permutations, with their induced permutations of the pair rows, act on its matching variables. A hypothetical irreducible display and every image under this group satisfy every matching clause and every reduction clause: each reduction is certified independently of the lex constraints. The lexicographically least image therefore satisfies both the reduction clauses and the inclusive lex comparisons; the reduction-clause list need

not itself be invariant. Independent input reconstruction and complete clausal refutations exclude every target in the table. The lower certificates and corresponding exact upper-bound packets are indexed in Appendix A. The four-column recursive values were already known from [2]; here the new conclusion is their actual-SOS optimality. □

The value $z_2(10, 5) = 34$ also answers the five-column case of [1, Section 8, Open Problem 1].

7 A uniform six-column construction

The remaining part of Theorem 1.4 is a construction for every $m \geq 15$. All coordinates in the following recipes are zero-based. Start from the 15×6 witness in [3, Figure 4], preserving its 30 singleton cells. Replace four of its selected pairs by chains:

Track	Left cell	Right cell	Outgoing/incoming columns	Singleton column
0	$(10, 3)$	$(12, 4)$	$(3, 4)$	5
1	$(0, 2)$	$(8, 5)$	$(2, 5)$	1
2	$(4, 0)$	$(13, 2)$	$(0, 2)$	3
3	$(10, 1)$	$(11, 4)$	$(1, 4)$	0

Write these columns as (a_i, b_i) and the new row on track i in block k as $u_i(k) = 15+4(k-1)+i$. With $u_i(0)$ and $u_i(t+1)$ the actual endpoint rows, select

$$Q_i(k) = \{(u_i(k), a_i), (u_i(k+1), b_i)\}, \qquad 0 \leq k \leq t. \tag{23}$$

In each block, place its four singletons at the listed columns and same-column bridges between track pairs $01, 02, 03, 12, 13, 23$ in columns $0, 1, 2, 4, 3, 5$, respectively. The normalized block is

	0	1	2	3	4	5
0	B_{01}	B_{02}	B_{03}	$Q_0(k)$	$Q_0(k-1)$	S_0
1	B_{01}	S_1	$Q_1(k)$	B_{13}	B_{12}	$Q_1(k-1)$
2	$Q_2(k)$	B_{02}	$Q_2(k-1)$	S_2	B_{12}	B_{23}
3	S_3	$Q_3(k)$	B_{03}	B_{13}	$Q_3(k-1)$	B_{23}

(24)

where bridge and singleton indices k are suppressed. Every cell is occupied exactly once, and every added singleton row has degree one.

7.1 Uniform certificate principles

We make explicit what permits a finite certificate to establish an arbitrary-length family. A chain whose consecutive pairs use fixed outgoing and incoming columns has the following wave property: once one selected pair is identified, its representative in the outgoing column grounds the companion of the next selected diagonal. One rectangle identifies that diagonal. Reverse propagation uses the incoming column. At either end the argument uses the actual core cell, not a fictitious extra module row.

A finite collar consists of the core and a fixed number of first and last blocks. A cell paired with one outside the collar retains its full-family class and is not treated as a hole.

Lemma 7.1 (Lifting symbolic certificates)**.** *Consider a family of finite displays indexed by an integer $t \geq t_0$. A finite list of record templates certifies the family if the following conditions hold on every template's entire integer domain:*

(i) Each record specifies cells and classes by affine integer expressions in t and local indices. Cell expressions evaluate to cells of the display, and class indices refer to its prescribed classes. Occupancy, membership, coincidences and distinctions hold throughout the domain. Every rectangle used has distinct rows and columns.

(ii) Each instantiated record is a valid line, hole, rectangle, complementary-pair or saturation inference from Section 2, or an application of an already proved lemma. Each noninitial premise is the conclusion of an earlier instantiated record in an explicit well-founded order.

(iii) The conclusions cover all selected-pair identifications and all orthogonality relations between distinct final classes.

Then every display in the family satisfies $\mathrm{RW3}^{+}$.

Proof. Fix $t \geq t_0$. Condition (i) makes every instantiated record a statement about this finite display. Well-founded induction using (ii) proves each record from the coefficient rules or the cited lemma. Condition (iii) gives the terminal conditions of $\mathrm{RW3}^{+}$. Since t was arbitrary, the conclusion holds throughout the family. □

In the certificates below, the domain checks are exact integer equalities, inequalities and case splits. The orders are given by successive normalization stages, decreasing distances between indices, or distances to a verified boundary. For mixed products, a graph vertex specifies a core class and a module type; an edge specifies a genuine rectangle and an integer shift of the module index. A path terminates at a known ground or an already certified boundary class. A closed walk with nonzero total shift can be iterated only after checking that every exit reaches such a boundary. The written recurrences below handle the exceptional components; the remaining record tables and domain checks are in the companion. This is a symbolic computer-assisted proof, not an inference from a finite sample of lengths.

7.2 Normalization and all-index orthogonality

For the $15 + 4t$ residue, four core/end-block certificates initiate the chains in order $1, 2, 3, 0$, using 44, 28, 13 and 19 primitive facts. The wave argument completes each track before it is used in the next stage. A further 118-fact certificate identifies the 26 retained core pairs. Bridges identify by the column rule. These windows are constant as coefficient systems for all $t \geq 3$.

After normalization, seven short symbolic rectangle words establish the boundary-chain families

$$Q_1(0) \perp Q_0(k), \quad Q_0(t) \perp Q_2(k) \quad (0 \leq k \leq t), \qquad Q_1(0) \perp Q_3(t), \quad Q_2(t) \perp Q_3(t).$$

They terminate at actual line grounds and use no core orthogonality conclusion. The module certificate then combines inward index transfers, near-diagonal windows, boundary/interior paths and a two-end window. Its verified domains cover every pair of module classes for every $t \geq 17$. The inward steps strictly decrease the distance between block indices; the bounded-distance and endpoint cases are separate finite templates.

There are 56 old nonport classes and 14 module phase types. Of the 784 mixed states, 782 have paths of at most ten transfers, with block shifts in $[-2, 3]$, to a line or a previously established module ground. These paths apply for $3 \leq k \leq t - 3$. The two exceptions have a common recurrence. Write $F = v_{1,0}$ for the singleton coefficient vector and $E = v_{1,1} = v_{1,4}$ for the normalized coefficient vector of the retained core pair. With

$$X_k = \langle F, Q_3(k) \rangle, \qquad Y_k = \langle E, S_3(k) \rangle,$$

a fixed three-rectangle word gives $X_0 = 0$, and the block gives

$$X_{k-1} = -Y_k = X_k \qquad (1 \le k \le t). \tag{25}$$

Hence both entire exceptional families vanish. A core plus first-five and last-five block patch covers the remaining old/old and boundary targets. Its 6,282 facts cover 5,684 required targets; its affine geometry is valid for $t \ge 11$. For $t \ge 17$ the interior and boundary ranges cover every index. Irrelevant cut-pair variables are not assumed zero. Every one of the 17 shorter lengths is checked separately.

7.3 The other three residues

For the $17 + 4t$ family, add two core rows with singleton column 0 and the five pairs

$$(15,1)(15,2), \quad (15,3)(15,4), \quad (16,1)(16,5), \quad (15,5)(16,3), \quad (16,2)(16,4).$$

The module now starts after row 16, with the same original four ports. No added pair crosses to the old core. The two-row subsystem has its own seven-form certificate. The enlarged mixed graph has 882 states, 880 ordinary paths and exactly the same two exceptions (25). Its core/boundary patch and the same index coverage complete this residue.

To either odd residue, attach one row with singleton in column 1, a hole in column 0, and pairs in columns $(2,5)$ and $(3,4)$. The finite boundary certificates move this last row to core position 15 or 17 and shift the module rows accordingly. This row permutation preserves every selected support; it is not an additional attachment. For the $16 + 4t$ family, 40 intrinsic phase paths ground the ordinary new/old products. If the new singleton is F' and its $(2,5)$ pair is A', the two exceptions satisfy

$$\langle F', Q_1(k-1)\rangle = -\langle A', S_1(k)\rangle = \langle F', Q_1(k)\rangle.$$

The actual right endpoint supplies a seed through the retained pair with cells $(8,1),(9,2)$, which shares column 2 with A'. A boundary patch uses this seed and only already proved facts of the $15 + 4t$ family, so the argument is not circular.

For $18+4t$, the $17+4t$ subsystem, the $16+4t$ subsystem, and the fixed three-row extra subsystem jointly cover every selected class pair. No selected support is cut, so their certificates embed. The four residue counts are

m	$\lvert F\rvert$	$\lvert D\rvert$	H	N
$15 + 4t$	$30 + 4t$	$30 + 10t$	0	$60 + 14t$
$16 + 4t$	$31 + 4t$	$32 + 10t$	1	$63 + 14t$
$17 + 4t$	$32 + 4t$	$35 + 10t$	0	$67 + 14t$
$18 + 4t$	$33 + 4t$	$37 + 10t$	1	$70 + 14t$

The uniform proofs apply for $t \ge 17$, and the 68 shorter configurations are covered by finite certificates. In every case $|F| = m+15$ and the display attains $U_6(m)$. This proves the infinite part of Theorem 1.4.

For $m = 2, 3, 4$, transposition and the prior two-column identity, three-column theorem and 6×4 equality give $7, 12, 16$ [8]. Theorem 6.1 supplies $m = 5, 6, 7$. For $8 \le m \le 14$, [3, Corollary 4.2] gives the actual cap, which, by Lemma 2.2, simplifies to $\lfloor (15m + 15)/4 \rfloor$. All parts of Theorem 1.4 now follow.

The least eventual threshold for equality to the cell cap $U(m, 6)$ is 8: at $m = 7$ the common value is 28 and the cap is 30. The formula $\lfloor (7m + 15)/2 \rfloor$ starts at $m = 14$, where the value is 56; at $m = 13$ it gives 53 instead of 52.

8 The odd-width families

The width-five formula in Theorem 1.5 follows from [13, Theorem 5.4 and Corollary 5.5] and the parameter hierarchy. The remaining three families share a one-row module. Their finite interfaces are interpreted using the certificate principles of Section 7.1.

A related distance-reducing rectangle argument is given in [13, Lemma 4.2].

Lemma 8.1 (Parallel one-row chains)**.** *In real module rows $k = 1, \dots, t$, $t \geq 1$, place $Q_i(k+1)$ in column a_i and $Q_i(k)$ in column b_i, for disjoint column pairs (a_i, b_i). An optional singleton $S(k)$ occupies one further column. The first and last classes have their other representatives in the core, in columns a_i and b_i, respectively. After the target tracks normalize, all distinct module classes are recursively orthogonal.*

Proof. Classes on the same track share a column. For distinct tracks, put $U(q,p) = \langle Q_i(q), Q_j(p)\rangle$. Equal or adjacent indices have a common module row; the case $q = p = t+1$ uses outgoing representatives in row t. If $q > p+1$, an outgoing/incoming rectangle gives

$$U(q,p) \longleftrightarrow U(p+1, q-1), \tag{26}$$

reducing the absolute gap by two. The symmetric move handles $p > q+1$. The larger index uses its outgoing representative first, so every row lies in $1, \dots, t$, even at the upper endpoint. For $Q_i(q)S(p)$, the row grounds are $q = p, p+1$; an incoming swap handles $q < p$, followed where needed by the same gap reduction. Distinct singleton classes share a column. Only tracks appearing in the target are needed, so the lemma is available during staged normalization without presuming other tracks. □

8.1 Widths seven and nine

For widths seven and nine, the cores and new row patterns are as follows. An entry (a_i, b_i) specifies the outgoing/incoming columns, and s the singleton column.

n	Core rows	Core $(\lvert F\rvert, \lvert D\rvert, H)$	s	Track columns	N after t rows
7	25	$(46, 64, 1)$	0	$(1,2), (3,4), (5,6)$	$110 + 4t$
9	36	$(72, 126, 0)$	0	$(3,5), (2,4), (6,1), (8,7)$	$198 + 5t$

The full core lists and port indices are in the corresponding data packets. Replacing each port by its chain preserves every other pair. Each new row is full and has one singleton, so the bases attain $m + \binom{n}{2}$ and their hole counts stay $1, 0$. The resulting lengths are the width-seven and width-nine rows of Theorem 1.5.

For $t \geq 5$, retain the core and the first two and last two module rows. The cut classes $Q_i(3)$ and $Q_i(t-1)$ are distinct and occupied. Staged finite certificates, with Lemma 8.1 used only after its target tracks normalize, identify the whole core and all chains. The remaining mixed graphs have the following complete counts.

n	Core-represented classes	Mixed vertices	Components
7	113	452	20
9	202	1,010	30

Splitting a port gives its two cells distinct classes, both with core representatives. At both widths, every component has a grounded spanning tree. The width-nine collar uses 64 staged

transports, each on two consecutive real rows with verified net shift one and only previously normalized target tracks. The resulting repetitions are valid throughout the module.

Each lifted mixed path either reaches its declared ground or reaches an actual first/last chain class, where the established core clique or the normalized diagonal prescription applies. The affine collar checks exclude accidental coincidences of omitted mates over the full domain $t \geq 5$, including its boundary $t = 5$. At each width, the five cases $0 \leq t < 5$ are separately certified.

At width seven, separate actual-cell certificates cover rows 19 through 24. Their maximum singleton counts are 39, 40, 42, 43, 44 and 45, and their displayed lengths are $4m+10$. Ordinary optimality follows from $2|F| - 3m \leq 21$ at row 19, the strict subterminal bound of Lemma 2.1 at row 20, and the terminal formula thereafter. Together with the uniform family from row 25, this proves the claimed range. At row 18 the degree bound gives $|F| \leq 37$ and hence $z_2(18, 7) \leq 81 < 82$, proving the least linear threshold 19. At width nine the new 36-row core starts the uniform family directly. At row 35, Lemma 2.1 gives $|F| \leq 70$ and therefore $z_2(35, 9) \leq 192 < 193$. Its least linear threshold is consequently 36.

8.2 Eleven columns

The direct 55×11 core has 110 singletons, 247 pairs and the single hole $(19, 9)$. The outgoing/incoming column pairs of $Q_0, \ldots, Q_4$ are $(1, 6), (8, 3), (7, 9), (5, 10), (4, 2)$, with singleton column 0. Adding t one-row modules gives $|F| = 110 + t$, $|D| = 247 + 5t$ and $N = 357 + 6t = 6m + 27$ at $m = 55 + t$.

The staged collar preserves the genuine hole and all occupied cut mates. The staged proof uses 66 unit-shift transports and establishes normalization and the 362-class source clique before mixed closure. Twenty-six mixed components have direct grounds. The remaining two use the normalized coefficient vectors

$$F_3 = v_{6,0}, \quad D_3 = v_{6,5} = v_{6,10},$$
$$F_4 = v_{7,0}, \quad D_4 = v_{7,2} = v_{7,4}.$$

Here each D_j is the class of an actual same-row selected pair. Rectangles in core row 6 or 7 and the real module row $54 + k$ give

$$\langle F_j, Q_j(k) \rangle = -\langle D_j, S(k) \rangle = \langle F_j, Q_j(k+1) \rangle, \qquad j = 3, 4, \quad 1 \leq k \leq t. \tag{27}$$

The initial classes are $Q_3(1) = v_{11,5}$ and $Q_4(1) = v_{43,4}$. They are distinct from F_3, F_4, respectively, so the previously proved source clique supplies both zero seeds. Induction exhausts these two components, including the terminal chain index $t+1$. These recurrences handle the two exceptional components. Exact affine checks cover all $t \geq 5$, and the five shorter cases are separately certified. The maximum base and cell cap prove equality for every $m \geq 55$. Two additional full displays at $m = 53, 54$ have $(|F|, |D|, H) = (107, 238, 0), (108, 243, 0)$ and lengths $345, 351$. Their bases consist of one triple row and the 52 pair rows not covered by that triple, with one extra singleton row at $m = 54$. Their pair budgets give $2|F| - 3m \leq 55$, proving ordinary optimality. Explicit coefficient certificates establish $\mathrm{RW3}^+$ for both displays and bridge the range to the uniform family from $m = 55$. At $m = 52$, the same bound gives $|F| \leq 105$ and hence $z_2(52, 11) \leq 338 < 339 = 6 \cdot 52 + 27$. Thus 53 is the least threshold. Packets O11 and O11_BRIDGE contain the uniform proof and the two finite certificates.

9 Signed exclusions and small-width equality

9.1 Four columns

Proposition 9.1. *One has* $z_{\mathrm{RL}}(11, 4) = z_{\mathrm{SL}}(11, 4) = 29$ *and* $z_{\mathrm{RL}}(12, 4) = z_{\mathrm{SL}}(12, 4) = 32$.

Proof. For $m \geq 6$, pair counting gives $z(m,4) = m+6$. Equality forces six rows containing the six distinct column pairs and $m-6$ singleton rows. The nonincreasing quadruples of anchor multiplicities classify all maximum bases up to row and column permutations. There are six types at eleven rows and nine at twelve rows. The respective cell caps are 30 and 33. Complete signed enumeration excludes both targets:

Grid	Maximum-base types	Target pairs	Search states
11×4	6	13	189,363,352
12×4	9	15	379,946,285

Every base is exhausted, with one hole in the eleven-row target and none in the twelve-row target. The per-base inputs and counts are recorded in S4.

The search branches on a remaining cell: either it is a hole if the remaining budget permits, or it is paired with each possible partner. Failed signed subfamilies are excluded by complete-pair heredity. Deleting a complete displayed form grounds all rectangle deviations incident with its removed cells, so the signed derivation on retained cells replays. A matching-component bound rejects only insufficient remaining capacity. The only symmetry reduction swaps two wholly untouched rows with identical fixed supports, while fixing the pivot, selected pairs and chosen holes. Thus every branch omission preserves a representative of every completion.

The independent reader matches the node and depth counts of the archived signed union-find search. It instead rebuilds a parity graph, colors its components by breadth-first search, and grounds components containing a zero, an odd cycle, or opposite selected inequalities. Only a grounded selected-pair deviation enables its identification equations. The algorithms use the signed criterion throughout; a recursive-line rejection is never a pruning condition. The source audit, complete inputs, calibration cases and termination logs are in packet S4.

The recursive lower values 29 and 32 were already obtained in [2]. Explicit twelve- and fourteen-pair witnesses are also replayed by a separate actual-cell rule reader in S4. The hierarchy gives the asserted equalities. These are signed upper bounds. The actual-SOS intervals remain $29 \leq z_2(11,4) \leq 30$ and $32 \leq z_2(12,4) \leq 33$. □

Corollary 9.2. *For every $m \geq 1$ other than possibly $m = 14$, $z_{\mathrm{RL}}(m,4) = z_{\mathrm{SL}}(m,4)$.*

Proof. At one row every cell is fixed. The exact values for two through eight rows are in [8, 14]; Theorem 6.3 covers nine and ten rows, and Proposition 9.1 covers eleven and twelve. At thirteen rows and every $m \geq 15$, the recursive constructions in [2, Table 1 and Theorem 1.1] attain the common cell cap. At fourteen rows, the remaining interval is $37 \leq z_{\mathrm{RL}} \leq z_{\mathrm{SL}} \leq 38$. □

9.2 Signed exclusions at 11×5 and 8×7

Theorem 9.3. *One has*

$$z_{\mathrm{RL}}(11,5) = z_{\mathrm{SL}}(11,5) = 37, \qquad z_{\mathrm{RL}}(8,7) = z_{\mathrm{SL}}(8,7) = 38.$$

Lemma 9.4. *One has $z(11,5) = 21$, and the maximum base is unique up to row and column permutations: ten rows carry the ten column pairs once each, and one row is a singleton. One has $z(8,7) = 22$, and there are exactly two classes of maximum bases:*

- *type A replaces one triple of a Fano triple system by two of its three pairs;*
- *type B adds one singleton row to a Fano triple system.*

Proof. With row degrees d_r, C_4-freeness gives $\sum_r \binom{d_r}{2} \leq \binom{n}{2}$.

- 11×5. Since $11 \geq \binom{5}{2}$, Lemma 2.1 gives $|F| \leq 21$ and forces ten rows of degree two, one on each column pair, and one row of degree one. Column permutations are transitive on the singleton's column.
- 8×7. The inequality $2d \leq 3 + \binom{d}{2}$ gives $2|F| \leq 24 + 21$, so $|F| \leq 22$. At $|F| = 22$, $\sum_r (d_r - 2)(d_r - 3)/2 \leq 1$, so every row has degree two or three, with at most one row of degree one or four. This leaves the degree profiles $2^2\,3^6$, $1\,3^7$ and $2^3\,3^4\,4$.

In the last profile, each triple meets the degree-four row in at most one column, since the base has no C_4. So each of the four degree-three rows contains at least two of the three columns outside the degree-four row, and hence a pair of them. Two triples share at most one column, so these four pairs are distinct. Only three such pairs exist, so this profile is impossible.

For $1\,3^7$, the seven triples cover all 21 column pairs, so they form a Fano system. The remaining row is a singleton, which gives type B.

For $2^2\,3^6$, the three column pairs missed by the six triples have even degree at every point, so they form a triangle. Adding that triangle completes a Fano system, and the two pair rows are two of its edges, which gives type A.

The Fano system is unique up to relabelling: its three triples through one point may be labelled $012, 034, 056$; choosing the triple through $1, 3$ as 135 forces the remaining triples $146, 236, 245$. Its automorphism group is transitive on points and on incident point–triple flags. Hence each type is a single class. □

Proof of Theorem 9.3. Lower bounds. Explicit recursive-line displays of lengths 37 and 38 on maximum bases were checked by the reference closure and an independent C++ closure in the recorded audit. The portable replay in packet X115/X87 rechecks them with the reference closure.

Reduction to hole-free displays. We have

$$U(11, 5) = \lfloor (55 + 21)/2 \rfloor = 38, \qquad U(8, 7) = \lfloor (56 + 22)/2 \rfloor = 39.$$

In both cases $mn + z(m, n)$ is even. So by the cell count $mn = |F| + 2|D| + H$, a display of length U has no hole. It therefore suffices to show that none of the three maximum bases of Lemma 9.4 carries a hole-free display satisfying RW3$^{\pm}$.

Heredity. Suppose (F, D) satisfies RW3$^{\pm}$ and $S \subseteq D$. Then the display (F, S), in which the cells of $D \setminus S$ become holes, also satisfies RW3$^{\pm}$ (Lemma 2.3). We use this in two ways:

- every pair of a signed display is a *candidate*, that is, a pair whose one-pair display is accepted;
- a set S of candidates whose display (F, S) fails RW3$^{\pm}$ cannot lie in a signed display. Such an S is a *cut*.

Region domains. Fix a set R of rows:

- at 11×5, the four pair rows through the singleton's column;
- on type A, its two pair rows;
- on type B, the three triples through the singleton's column.

The *pattern* of a display is its set of pairs with both cells in R. By heredity it is itself an accepted display on candidates. An exhaustive enumeration of all partial matchings of the free cells of R lists every such pattern. Every accepted pattern (4,968, 1,449 and 353 of them) is mapped onto one of 229, 190 and 18 listed representatives by a base automorphism preserving R. Each map is checked to preserve the base.

Propositional encoding. For a base, take one variable x_P for each candidate pair P, and the following clauses:

- exactly one selected pair meets each free cell;
- for every cut S found, the clause $\bigvee_{P \in S} \neg x_P$.

Call this formula B. For each representative d, let L_d be the unit literals fixing the region pattern to d. A hole-free signed display, moved by an automorphism so that its pattern is a representative, satisfies $B \wedge L_d$ for that d. The formulas contain no symmetry-breaking clauses; the identical-row swap of Proposition 9.1 has no instance on these bases. Hence it suffices to refute $B \wedge L_d$ for every representative d. The final formula adds *selector* clauses: a new variable s_d for each d, the clauses $s_d \Rightarrow L_d$, and $\bigvee_d s_d$.

The cuts were generated by counterexample-guided search. Their provenance is irrelevant to the proof, because every cut is re-checked.

Refutations. Each refuted formula is a subset of $B \wedge L_d$: the unit clauses L_d, the propagating clauses, and the clauses of B not satisfied by unit propagation from L_d. The checker verifies this subset relation line by line.

For a few domains the refutation is split on the partner of one free cell c outside R. Since B contains the clause $\bigvee_{P \ni c} x_P$, refuting $B \wedge L_d \wedge x_P$ for every candidate $P \ni c$ refutes $B \wedge L_d$.

Every refutation is a CaDiCaL LRAT proof. It is accepted by cake_lpr, a formally verified checker [10], and separately by the LRAT checker of the drat-trim distribution.

Audit. Every clause of each final formula is classified:

- exactly-one clauses are rebuilt from the candidate list, and selectors from the representative list;
- every other clause must be a negative clause on candidates, and its pair set is re-checked as a failed display, by the C++ closure and by the reference closure;
- every excluded non-candidate pair is re-checked;
- the region literals are rebuilt from the representatives.

The coverage enumeration and the candidate/non-candidate split were repeated with the reference closure. In addition, a random sample of 34 domains and split cases, at least eight for each base, was refuted again by kissat, and drat-trim accepted all 34 DRAT proofs.

The counts are as follows.

Base	Candidates	Domains	Cut clauses	Split domains	Split cell cases
11×5	387	229	5,168,225	4	4×21
8×7 type A	326	190	946,334	1	16
8×7 type B	312	18	589,720	1	15

The cut counts refer to the final formula of each base. At 11×5 and on type A, 213 and 189 domains respectively were refuted from an earlier formula of the same kind, which was audited in the same way (3,956,309 and 946,270 cut clauses).

No hole-free signed display of length $U(m, n)$ therefore exists, so $z_{\mathrm{SL}}(11, 5) \leq 37$ and $z_{\mathrm{SL}}(8, 7) \leq 38$. The hierarchy completes the proof. □

Remark 9.5. These are upper bounds for the signed criterion, not for the actual square length. The actual values remain in the intervals $37 \leq z_2(11, 5) \leq 38$ and $38 \leq z_2(8, 7) \leq 39$.

Together with the other finite values and the width-five formula of Xi and Chang [13], Theorem 9.3 completes the width-five comparison (Corollary 9.7). At 8×7 the common value 38 lies one below the cell cap $U(8, 7) = 39$.

At 14×4 the remaining interval $37 \leq z_{\mathrm{RL}}(14, 4) \leq z_{\mathrm{SL}}(14, 4) \leq 38$ is the subject of Problem 5.1 in [2].

Proposition 9.6. *One has* $z_{\mathrm{RL}}(9, 5) = z_{\mathrm{SL}}(9, 5) = z_2(9, 5) = 31$.

Proof. Lemma 2.1 gives $z(9,5) \le 18$. The following display has 18 singleton cells and 13 pairs, and its singleton set is C_4-free. Equal numbers form one selected pair; bullets are singletons and the circle is a hole. Packet BOUNDARY checks its recursive-line closure.

	0	1	2	3	4
0	•	•	1	4	5
1	•	3	•	7	4
2	•	1	2	•	9
3	•	6	8	12	•
4	9	•	•	12	10
5	11	•	13	•	2
6	13	•	11	8	•
7	10	5	•	•	○
8	6	7	•	3	•

The base is maximum and $U(9,5) = \lfloor (45+18)/2 \rfloor = 31$, so the hierarchy proves the assertion. □

Corollary 9.7. *For every $m \ge 1$, $z_{\mathrm{RL}}(m,5) = z_{\mathrm{SL}}(m,5)$. Consequently $z_{\mathrm{RL}}(m,n) = z_{\mathrm{SL}}(m,n)$ for every rectangle with $\min(m,n) \le 6$, except possibly 14×4 and 4×14.*

Proof. Both criteria are invariant under transposition. At width five, rows one through four follow from the one-, two- and three-column values and the 5×4 equality in [8, 14]. At five rows the common value is 17 [8, 14]; at six rows it is 20 by Theorem 6.1 and transposition. Theorem 6.3 gives $24, 27, 34$ at rows seven, eight and ten. Proposition 9.6 gives 31 at nine rows. Eleven rows are covered by Theorem 9.3, and every $m \ge 12$ by Theorem 1.5. Widths one to three are in [8], width four is Corollary 9.2, and width six is Theorem 1.4. This covers every rectangle with a side at most six, with the stated exception. □

9.3 A fixed-base gap

Proposition 9.8. *There is a maximum 7×5 base F with $R(F) = 8$ and $S(F) = 9$, although $z_{\mathrm{RL}}(7,5) = z_{\mathrm{SL}}(7,5) = 24$.*

Proof. Take the 7×5 base with row supports

$$012, \quad 034, \quad 13, \quad 14, \quad 23, \quad 24, \quad 1.$$

It has 15 singletons and is ordinary maximum by pair counting. Figure 1 displays an eight-pair recursive witness and a nine-pair signed witness; packet `FIXED_BASE_NINE` contains their certificates.

A complete recursive-line exclusion fixes each of the 190 possible two-hole sets and exhausts every pairing of the other 18 free cells. Only rejection of a recursive-line subfamily prunes the search, which is valid by complete-pair heredity. An independent physical-cell reader exhausts 764,150 prefixes and 2,617,789 closure calls, finding no nine-pair recursive display. Every larger recursive display would contain such a subfamily. Thus $R(F) = 8$. The signed lower witness and the actual bound $z_2(7,5) = 24$ prove $S(F) = 9$. Another maximum base at 7×5 admits nine recursive pairs, so this example is consistent with the global equality in Theorem 6.3.

□

	0	1	2	3	4
0	•	•	•	6	○
1	•	○	7	•	•
2	○	•	○	•	5
3	5	•	4	1	•
4	3	7	•	•	6
5	4	1	•	2	•
6	2	•	8	8	3

	0	1	2	3	4
0	•	•	•	7	○
1	•	1	3	•	•
2	3	•	4	•	7
3	○	•	6	5	•
4	5	2	•	•	1
5	6	8	•	8	•
6	2	•	9	9	4

Figure 1: The same maximum singleton base at 7×5. Left: eight recursive-line pairs. Right: nine signed pairs. Each bullet is a distinct singleton, each circle is a hole, and equal numbers within one grid form a selected pair. The right grid is signed admissible; no recursive-line display on this base has nine pairs.

A Certificates and verification scope

The manuscript gives the structural proofs and mathematical reductions for the computer-assisted results. The infinite exact families also use external symbolic record tables satisfying Lemma 7.1; the finite tables and their exact domain checks are essential supporting evidence. The component map is:

Component	Claims supported
FINITE_SIX	Theorem 6.1: actual six-column upper bounds
SIX_FAMILY	All-length six-column construction
FINITE_FIVE, FINITE_FOUR	Theorem 6.3
F7	The 7×7 equality
S4	Signed exclusions at 11×4 and 12×4
BOUNDARY	The 9×5 witness
FIXED_BASE_NINE	Proposition 9.8
F16	Proposition 5.7
O5_EXTENSION	Shared one-row certificate readers for O7
O7, O9	Odd-width families at seven and nine
O11, O11_BRIDGE	Width eleven and its two boundary cases
X115/X87	Theorem 9.3

The finite SOS upper bounds use explicit identities and checked clausal refutations. The S4 exclusions and the fixed-base recursive exclusion use independent exhaustive implementations rather than separate clausal refutations; reproducing those bounds requires rerunning the enumerations. Signed exclusions use failed signed subdisplays and Lemma 2.3; they are not actual-SOS exclusions. X115/X87 contains the formulas, domain representatives, coverage records and complete audit receipts described in Section 9.2. Its readers regenerate LRAT proofs rather than storing all traces.

The supporting archives are deposited on Zenodo at `https://doi.org/10.5281/zenodo.23124764`.

The main companion is `focused_evidence.zip`, with SHA-256

`f68d42718c764764d27233ab6e0ad5d0411a2aeeaab5a3de8998c44104f6c714`

The X115/X87 packet is `signed_exclusions_evidence.zip`, with SHA-256

`bb43da8af5d939f2d476c6c74bf70730fcf1441a1c79ef6bc767659374663d53`

The README gives per-component replay commands, dependencies and tool requirements. Integrity checks verify bytes only; sampled replay is not full proof regeneration.

The general density, two-column extension and transfer-graph results have written proofs. Auxiliary finite checks do not establish their infinite ranges. Lean sources formalize the coefficient/graph equivalence in Theorem 5.4 and the existence of a nonzero product relation exactly when an ungrounded bipartite component exists. The formal statement uses occupied classes and vanishing for all real row and column vectors; the geometric bridge is proved by finite-sum identities. The dimension count, SOS rank consequence, closure algorithms and computational bounds remain outside the formalization. An abstract chain-propagation induction is also included; its geometric premises remain assumptions. The Lean toolchain and mathlib revision are pinned.

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AI assistance and responsibility. AI tools assisted exploratory computations, software development and draft preparation. The author is responsible for the mathematical claims, exposition and supporting data.

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